

FORENSIC ANALYSIS REPORT – JEAN’S DISK IMAGE (M57 CASE)

Forensic Artifact Recovery

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1. Introduction

The purpose of this forensic analysis was to examine the digital evidence contained within the disk image nps-2008-jean.E01, acquired from the publicly available dataset at Digital Corpora, <https://digitalcorpora.org/corpora/scenarios/m57-jean/>. This disk image represents the workstation (version: 25H2) of an employee named Jean at the fictional company M57.biz. The investigation followed the academic requirements outlined in Assignment_11_Forensic_Artifact_Recovery, focusing on identifying significant artifacts, reconstructing user activity, and assessing whether any actions or data found on the system may indicate security policy violations, mishandling of sensitive information, or potentially harmful activity.

To ensure ethical and responsible handling, all examinations were conducted in a controlled forensic environment, and the image itself was never modified. All extracted materials referenced in this report were saved to the project repository under artifacts/, and the chronological reconstruction of activity is preserved in timeline/timeline.csv. The overall intention of this report is to provide a clear and professional forensic interpretation of the events and data found on the disk, without speculation and while complying with accepted forensic methodology.

2. Evidence Integrity and Handling

From the moment the disk image was imported into Autopsy, evidence of integrity was maintained through strict observance to forensic principles. The image was opened in read-only mode, and no operations were performed that could alter the original data. All exported content, such as screenshots, metadata captures, and recovered documents, were saved into the designated artifacts/images/ directory to ensure a transparent and reproducible workflow. Metadata such as Created, Modified, Accessed, and Changed timestamps were preserved exactly as they appeared in the image, and the report reflects them without transformation or interpretation.

No hash mismatches or warnings were generated during the imaging or import process. This indicates that both the source material and the extracted artifacts remained intact throughout the investigation. The lab environment ensured that the examiner system could not introduce contamination, and the collected materials were stored in a separate, documented structure as part of the repository.

3. Overview of System Activity and File Population

The disk analysis revealed a workstation with significant user activity. Autopsy identified over 10,000 files, along with 261 email messages, 40 installed programs, 42 shell bags, 5,339 web history entries, and 238 search queries. This volume reflects a system used actively for communication, browsing, and document management. The presence of attached USB devices, stored cookies, downloaded files, and browser autofill entries provided a broad picture of Jean's typical workflow and digital habits.

Deleted files were also analyzed, but none of them contained meaningful or suspicious content, and they did not appear to contribute to an understanding of the case. As a result, they were not considered relevant for further investigation. The focus instead shifted toward artifacts that stood out either because of their content, their metadata, or their contextual significance when paired with other events on the timeline.

4. Rationale for File Selection

Given the large number of files present on the system, it was necessary to narrow the investigation into those artifacts that appeared unusual, sensitive, or potentially related to harmful or policy-violating activity. The selected files include images with firearm-related content, a document containing sensitive personnel information such as Social Security Numbers (SSNs), and a sequence of email messages that relate directly to the handling and possible exposure of confidential data.

These selections were made not because they inherently represent criminal behavior, but because they raised questions about user intent, observance to workplace policy, and the possibility of accidental data leakage. Forensic analysis prioritizes files that may establish motive, demonstrate negligence, or correlate with known incidents. In this case, the presence of a salary and SSN spreadsheet combined with emails about background checks and later emails discussing leaked SSNs clearly justified closer inspection. All files referenced are linked to the project repository, for example:

artifacts/images/salary_sheet.png,

artifacts/images/email_background_checks.png, and others included in the timeline.

5. Findings

5.1 Image Evidence

Three notable images were discovered during the examination. Two of these depicted a pistol and ammunition, both created on the evening of July 12, 2008. The metadata, visible in Autopsy and captured in artifacts/images/pistol.png and artifacts/images/ammo.png, shows that these files were created and modified within seconds of each other. A third image, artifacts/images/photo_2008_07_13.png, was created early the next morning.

While the presence of firearm images is not illegal, storing them on a work computer could violate internal policy and be considered inappropriate. The proximity of their timestamps also suggests the user may have imported or generated multiple related images during the same session. None of these images directly relate to the SSN data breach, but they do form part of the overall behavioral profile of the user and therefore warrant documentation.

5.2 Sensitive Document Containing Employee SSNs

One of the most significant findings was a spreadsheet titled M57.biz company, recovered and saved under artifacts/images/salary_sheet.png. This document contains detailed salary information, employee names, and Social Security Numbers. The metadata indicates that the file was created, accessed, and modified around 03:28 on July 20, 2008. Its presence on Jean's machine strongly correlates with the subsequent email exchanges, which suggest that real SSNs were used in a database that was not intended to store actual personal data.

The handling of this document appears to be at the center of the incident. The forensic timeline (timeline/timeline.csv) places its creation just one day before multiple colleagues contacted Jean expressing concern about leaked SSNs. This temporal link strengthens the interpretation that the spreadsheet was either shared improperly or integrated into a system that made it publicly accessible.

5.3 Email Evidence and Data Exposure

Email analysis revealed a sequence of messages that provide essential context to the data breach. Early messages created on July 6, 2008 (visible in artifacts/images/email_background_checks.png and corresponding files), show that Alison requested Jean to compile a background-check spreadsheet including sensitive information. This indicates that Jean was formally tasked with managing employee SSNs.

By July 21, several incoming emails suggest that SSNs have been publicly exposed. Bob's email, preserved as artifacts/images/email_ssn_question.png, directly asks Jean whether he is aware that Bob's own SSN has appeared online. Carol, the company's webmaster, later writes to Jean questioning whether Jean mistakenly populated the company's test database with real employee data. This message (artifacts/images/email_realdata.png) is critical

because it implies a technical pathway through which the leak could have occurred: a test environment not intended for real data was exposed to external access.

The most concerning message that Jean gets is Please send me the information now, found in artifacts/images/email_send_info.png, where Jean appears to transmit sensitive information without any encryption or security precautions. Taken together, these emails form a coherent narrative: Jean was asked to perform a sensitive task, mishandled the data, possibly uploaded real SSNs to an unsafe system, and colleagues later discovered traces of the leaked information.

An additional observation according to email_send_info.png concerns the sender information in the email exchange. The message requesting the information urgently is displayed as being sent from alison@m57.biz while the underlying sender address is listed as tuckgorge@gmail.com. This discrepancy indicates that the visible sender identity does not match the actual email address used to send the message. The use of a different sender address raises the possibility that the message was not sent by the intended individual, and that the identity of Alison may have been used in communication. Such mismatches between displayed sender information and actual email addresses are commonly associated with email spoofing and phishing-related attacks, where sender identities are manipulated during communication. The email contains a request for immediate action, which resulted in Jean sending the requested information without considering whether the message could have been malicious, fraudulent, or sent from a compromised account, she also sent the message without the use of encryption or additional security measures. This observation is relevant when assessing how the information was transferred and under what circumstances the data was disclosed.

6. Timeline Reconstruction

The activity timeline was collected and stored in timeline/timeline.csv provides the chronological backbone of the investigation. By aligning the timestamps of images, documents, and emails, a clear sequence emerges. System files and application libraries show normal activity until mid-2008. On July 12–13, Jean created several images. On July 20, the sensitive SSN spreadsheet was created. Within hours of this event, multiple emails on July 21 reflect concern from coworkers about potential data exposure. This clustering of timestamps strongly supports the conclusion that the data breach occurred between the creation of the spreadsheet and the subsequent email warnings.

7. Implementation of AI

Artificial Intelligence was used as a supportive tool to assist with structure and efficiency during the forensic workflow, without replacing forensic analysis or examiner judgment. The AI model used was Mistral 7B Instruct v0.3, and it was only provided with manually selected data extracted by the investigators.

One of the challenges of the project was identifying an accessible and suitable disk image. Therefore, Artificial Intelligence was used, based on instructions provided by us, to support the process of selecting a disk image that met the project requirements.

AI was primarily used to assist in the creation of the investigation timeline. The tool was intended to organize the selected artifacts chronologically based on the metadata from the artifacts which were considered most relevant to the case. The investigators still performed manual adjustments to ensure the correct order of events and to verify the timeline's accuracy. This approach reduced irrelevant data and allowed the timeline to reflect only the artifacts deemed significant for the investigation. A built-in timeline function exists in Autopsy but using it would have generated a timeline covering the entire disk, which contains over 10,000 files and would have produced a large amount of irrelevant data.

Additionally, AI was used to support documentation-related tasks. This includes structuring the report, organizing the GitHub repository, and formatting the timeline into a CSV file. AI was not given access to the disk image, Autopsy or raw evidence, and it did not perform any automated evidence of extraction or analysis. All AI-assisted output was reviewed, validated, and if necessary, modified to ensure accuracy and consistency with the project and forensic methodology.

8. Conclusion

The forensic investigation of Jean's workstation reveals a pattern of accidental data mishandling rather than intentional wrongdoing. Jean was assigned a task requiring the processing of sensitive employee information, including Social Security Numbers. Through a combination of user errors and likely misunderstanding of proper data handling procedures, this information appears to have been placed in an insecure environment. Colleagues quickly discovered the exposure, as reflected in their emailed concerns.

Although no evidence indicates malicious behavior, Jean's actions clearly violated basic principles of data protection. The presence of firearm images on the same system further demonstrates inconsistent or inappropriate personal use of work equipment, though these files do not appear to be directly linked to the SSN incident. The findings highlight the need for improved security training, stricter policies regarding sensitive data, and technical safeguards such as Data Loss Prevention tools.

9. Recommendations

M57.biz should consider implementing formal policies for handling personal data, including encryption requirements, access restrictions, and validated procedures for background checks. Test environments must never store real personal information, and they should implement training for employees that clearly define differences between safe and unsafe data-handling practices. Automated monitoring tools may also help prevent similar incidents by detecting sensitive data before it leaves secure domains.